

XRIM-117 WYVERN PTD — Dedicated Aerodynamic Ground Test Facility

Open-return low-speed wind tunnel · Full 3D-printed · Designed for 1:3 scale PTD PDR-002 Rev A test articles

Programme: PDR-002 Rev A Test scale: 1:3 geometric Fan: AC Infinity Cloudline T8 Contraction: 9:1

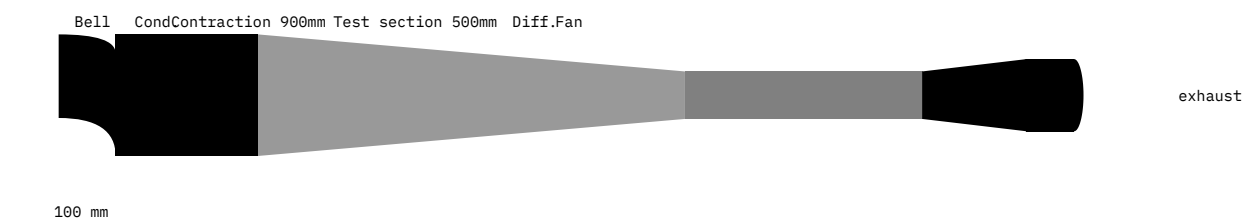
Test section: 150 × 150 × 500 mm OAL: 2,468 mm

Overview	Module specs	Aerodynamics	Contraction profile	Print plan	BOM & cost
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CRITICAL DIMENSIONS

Test section (H × W × L)	Test section area	Contraction ratio
150 × 150 × 500 mm	225 cm ²	9 : 1
Settling chamber	Test section velocity	Dynamic pressure q
450 × 450 mm	7.3 m/s (24 ft/s)	33.0 Pa
Fan	Flow rate (loaded est.)	Contraction cone length
AC Infinity T8 (200 mm)	350 CFM / 0.165 m ³ ·s ⁻¹	900 mm
Diffuser length	Full tunnel OAL	Wall thickness (all modules)
220 mm (4° half-angle)	2,468 mm (2.47 m)	3.0 mm PETG-CF
Module OD	Junction ring OD	Honeycomb cell
156 × 156 mm	176 mm	4 mm hex, L/d = 7
Screen set	Smoke ports	Bell-mouth lip radius
20-mesh + 80-mesh stainless	3 × 3 grid, 50 mm pitch	180 mm
Est. turbulence intensity	Total print mass	Estimated filament cost
~0.21 %	~4,109 g (4.1 kg)	~\$115 USD

TUNNEL LAYOUT — SIDE ELEVATION



BLOCKAGE SUMMARY (1:3 SCALE MODEL, 150 × 150 MM TEST SECTION)

TEST ARTICLE	FRONTAL AREA	BLOCKAGE	STATUS
Nose cone on sting (body only)	4.28 cm ²	1.9 %	No correction
Single fin panel — half-span wall mount	5.44 cm ²	2.4 %	No correction
Ring 2 assembly + body stub	6.97 cm ²	3.1 %	No correction
Ring 1 assembly + body stub	15.17 cm ²	6.7 %	Maskell correction
Full rocket, 0° AoA (both rings)	17.82 cm ²	7.9 %	Maskell correction
Full rocket, 45° roll	~23.6 cm ²	~10.5 %	Visualization only

The 150 × 150 mm test section resolves the primary limitation of the source STEM tunnel (80 × 130 mm, 37% blockage at 1:1). The full-rocket at 0° AoA now sits at 7.9% — within the Maskell-correctable range. At 1:3 scale, body OD = 23.3 mm, Ring 1 fin span = 23.3 mm, OAL = 390 mm.